Command to upload bitstream to fpga – i am using arty a35t

python3 esp32_fpga_program.py /dev/ttyUSB0 counter_slave/build/arty_35/counter_slave.bit

replace these accordingly:

/dev/ttyUSB0

counter_slave/build/arty_35/counter_slave.bit

And you must place the:

esp32_fpga_program.py

file to the xc7 directory

command to list usb devices:

ls -l /dev/ttyUSB* /dev/ttyACM* /dev/ttyS* 2>/dev/null

```
crw-rw---- 1 root dialout   4, 72 Aug 14 23:27  /dev/ttyS8
crw-rw---- 1 root dialout   4, 73 Aug 14 23:27  /dev/ttyS9
crw-rw---- 1 root dialout 188,  0 Aug 14 23:40  /dev/ttyUSB0
kalii@ubuntu:~$ ld
ld: no input files
kalii@ubuntu:~$ ls
conda_installer.sh  librelane-venv  Public                            snap
Desktop             mgc             python2.7                         Templates
Documents           minicom.log     Python-2.7.18                     transcript
Downloads           Music           Python-2.7.18.tgz                 Videos
f4pga-examples      opt             Questasim
librelane           Pictures        Scala-Chisel-Learning-Journey
kalii@ubuntu:~$ cd f4pga-examples
kalii@ubuntu:~/f4pga-examples/xc7$ python3 esp32_fpga_program.py /dev/ttyUSB0 counter_slave/build/arty_35/counter_slave.bit
====================================
ESP32 -> XC7A35T JTAG Programmer
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Reading bitstream...
Payload offset : 0x6B
Payload size   : 2192012 bytes

Opening /dev/ttyUSB0 at 921600 baud...

1. Resetting JTAG TAP...
2. Sending JPROGRAM...
3. Waiting for FPGA initialization...
4. Sending CFG_IN...
5. Sending FPGA bitstream...

Programming 2192012 bytes...
Programming: 100.00% Elapsed:    53.4s

Bitstream programming finished in 53.36 seconds

6. Sending JSTART...
7. Sending startup clocks...

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FPGA PROGRAMMING COMPLETE
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kalii@ubuntu:~/f4pga-examples/xc7$
```
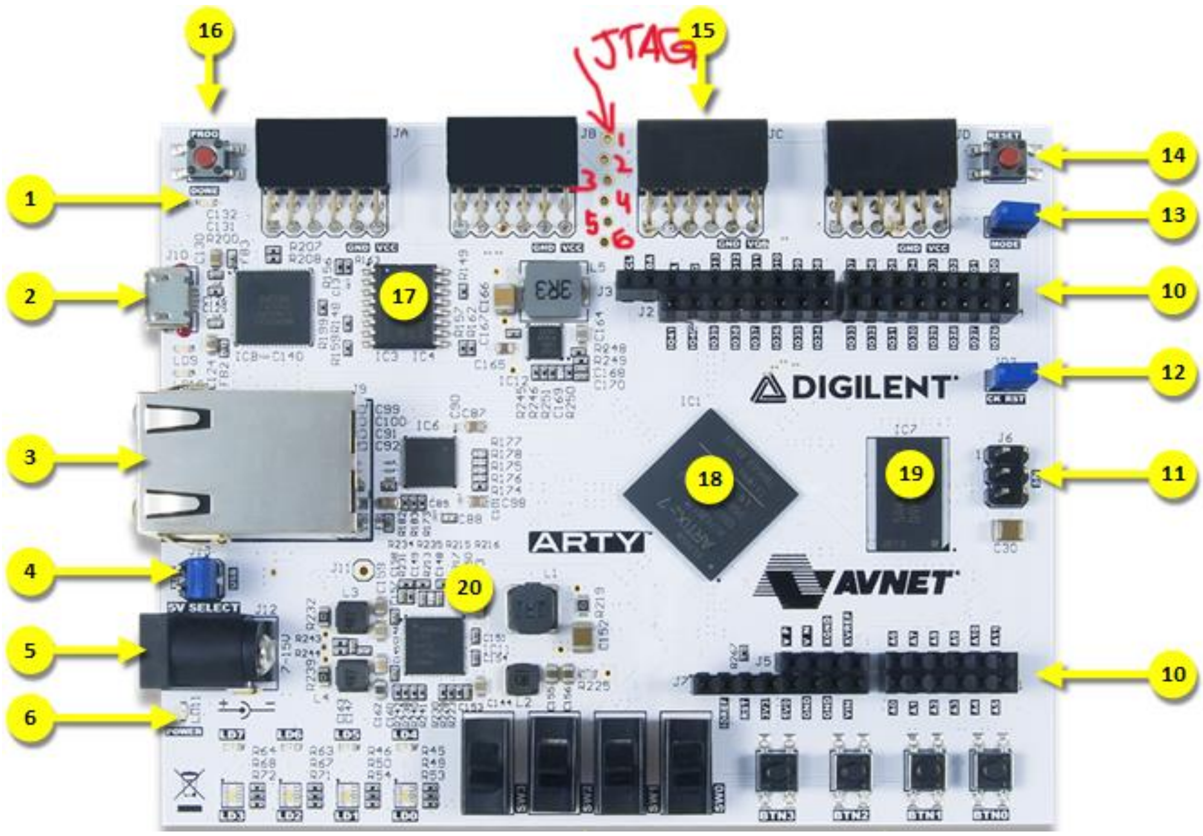

Connections Details:

The J8 port on the Digilent Arty (A7/S7) FPGA board is a 6-pin external programming header located on the underside of the board. Pin 1 is typically marked with a square pad or white indicator dot, mapping to the standard JTAG signal order:

Pinout Details

- **Pin 1:** TMS (Mode Select, input) (square pad)
- **Pin 2:** TDI (Data In, input)
- **Pin 3:** TDO (Data Out, output)
- **Pin 4:** TCK (Clock, input)
- **Pin 5:** GND (Ground)
- **Pin 6:** VCC (3.3V Power reference)



Connection with Esp32:

- GPIO 13 -> FPGA TCK
- GPIO 27 -> FPGA TDI
- GPIO 26 <- FPGA TDO
- GPIO 25 -> FPGA TMS
- GND <-> FPGA GND